

B.E. I EXAMINATION Jan' 2024
Information Technology
1AMRC1: Applied Mathematics-I

Time: 3 Hrs.

Max. Marks: 60

Note: Attempt any TWO parts from each Question. Each carries equal marks.

- Q1. (a) If $y = \sin(m \sin^{-1} x)$, prove that $(1 - x^2)y_{n+2} - 2(n+1)xy_{n+1} + (m^2 - n^2)y_n = 0$ and also find $(y_n)_0$. (06)
- (b) Use Taylor's series, to prove that $\tan^{-1}(x+h) = \tan^{-1} x + (h \sin z) \cdot \frac{\sin z}{1} - (h \sin z)^2 \cdot \frac{\sin 2z}{2} + (h \sin z)^3 \cdot \frac{\sin 3z}{3} - \dots$ (06)
- (c) Find the radius of curvature at the point 't' of the curve cycloid $x = a(t + \sin t), y = a(1 - \cos t)$. (06)
- Q2. (a) If $u = \sin^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$, prove that (06)
- (i) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$
- (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{-\sin u \cos 2u}{4 \cos^3 u}$
- (b) If $y_1 = r \sin \theta_1 \sin \theta_2, y_2 = r \sin \theta_1 \cos \theta_2, y_3 = r \cos \theta_1 \sin \theta_3, y_4 = r \cos \theta_1 \cos \theta_3$ then show that $J(y_1, y_2, y_3, y_4) = r^3 \sin \theta_1 \cos \theta_1$. (06)
- (c) Find all the maximum and minimum value of function f given by $f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$ (06)
- Q3. (a) Trace the curve $y^2(1 - x^2) = x^2(1 + x^2)$, writing steps properly. (06)
- (b) Find the whole length of the loop of the curve $3ay^2 = x(x - a)^2$. (06)
- (c) Find the area of the surface generated by the intercept $\frac{x}{a} + \frac{y}{b} = 1$ between the axis of coordinates, revolving about y-axis. (06)
- Q4. (a) Find by double integration the area enclosed by the pair of curves $y = 2 - x$ and $y^2 = 2(2 - x)$. (06)
- (b) Compute $\iiint \frac{dx dy dz}{(x+y+z+1)^3}$ if the region is bounded by the coordinate planes and the plane $x + y + z = 1$. (06)
- (c) Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy dx dy$ and hence evaluate it. (06)
- Q5. (a) Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, if vector F is: (06)
- (i) $F = (2y - \cos x)i - z^2 e^{3x}j + (x^2 + 7z)k$
- (ii) $F = z^2(y - x)i + \frac{4y^2}{z^3}j + (x^2 + 3z)k$
- (b) Verify Green's theorem in the plane for $\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the boundary of the region defined by $y = x^2, x = y^2$. (06)
- (c) State Stoke's theorem and evaluate for $F = (x^2 + y^2)i^* - 2xy j^*$ taken around the rectangle bounded by the lines $x = \pm a, y = 0$ and $y = b$. (06)
